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Practical 06

Aim : Cost Estimation of COCOMO Model

COCOMO Model is a procedural cost estimate model for Software Projects and is often used as a process of reliably predicting the various parameters associated with making a project such as size, effort, cost, time, and quality.

Types of Projects in COCOMO Model

1. **Organic**
2. **Semi-Detached**
3. **Embedded**

- **For the skill exchange website we will be using a semi-detached model ,since it's moderately complex but not as rigid as embedded systems.**

Code:

```
def cocomo_basic(kloc, mode="organic"):
    """
    Basic COCOMO Model
    kloc: estimated size of project in Kilo Lines of Code (KLOC)
    mode: 'organic', 'semi-detached', or 'embedded'
    """

       modes = {
        "organic": {"a": 2.4, "b": 1.05, "c": 2.5, "d": 0.38},
        "semi-detached": {"a": 3.0, "b": 1.12, "c": 2.5, "d": 0.35},
        "embedded": {"a": 3.6, "b": 1.20, "c": 2.5, "d": 0.32}
    }

    if mode not in modes:
```

```
        raise ValueError("Invalid mode. Choose 'organic', 'semi-detached', or 'embedded'.")

    params = modes[mode]

    effort = params["a"] * (kloc ** params["b"])

    time = params["c"] * (effort ** params["d"])

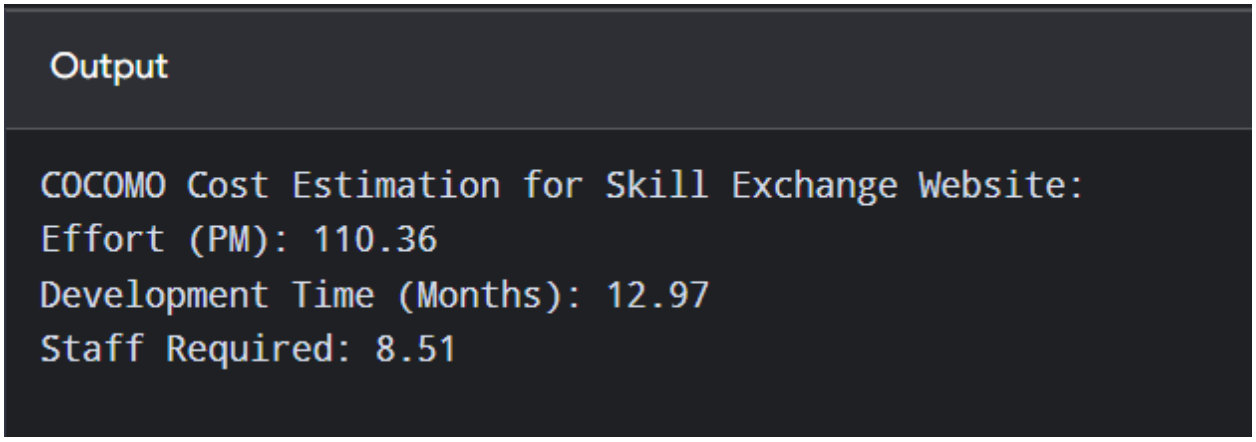
       staff = effort / time

    return {
        "Effort (PM)": round(effort, 2),
        "Development Time (Months)": round(time, 2),
        "Staff Required": round(staff, 2)
    }


kloc_estimate = 25  # Example: 25 KLOC estimated for your project
result = cocomo_basic(kloc_estimate, mode="semi-detached")

print("COCOMO Cost Estimation for Skill Exchange Website:")
for key, value in result.items():
    print(f"{key}: {value}")
```

Output:



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